The Valid Dictionary: Newtonian Mechanics

Can Valid

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Valid Dictionary (VD) Document: Newtonian Mechanics as a Valid Dictionary	**Status:** DRAFT
Internal ID: `vd-newton`	**Project tag:** `vd-project`

Point. Present a complete, minimal Newtonian mechanics encoding as a VD instance, demonstrating auditability and explicit structure.

Target goals.

- Provide a finished Newtonian VD with clear entry indexing and cross-references.
- Encode the relevant Newtonian "laws" as VD triplets consistent with `vd-core`.
- Keep commentary minimal: enough to read and verify the dictionary, not to persuade.

Non-goals.

- Arguing why the case study is important (belongs in `vd-newton-analysis`).
- Introducing new VD rules or changing the core spec.
- Covering advanced Newtonian topics beyond "basics" needed for demonstration.

Target audience. Readers who want to see VD used on real physics; skeptics who want something concrete to inspect.

Dependencies / references. Depends on `vd-core` for definitions, notation, and validity criteria.

1 Design 1

1.1 Non-law kinematics scaffold

E1 `time` := "A real parameter $t \in \mathbb{R}$ used to order events; differences Δt are durations measured by a clock."

E2 `reference_frame` := "A convention for assigning spatial coordinates and a time coordinate t to events: an origin, axes, and a clock."

E3 `position` := "In a reference_frame, an object's location at time t represented by a vector $\mathbf{r}(t) \in \mathbb{R}^3$."

E4 `displacement` := "$\Delta\mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1)$."

E5 `trajectory` := "The time-ordered curve $\{\mathbf{r}(t) : t \in I\}$ for some interval $I \subset \mathbb{R}$."

E6 `velocity` := "$\mathbf{v}(t) = d\mathbf{r}/dt$."

E7 `speed` := "$v(t) = \|\mathbf{v}(t)\|$."

E8 `acceleration` := "$\mathbf{a}(t) = d\mathbf{v}/dt = d^2\mathbf{r}/dt^2$."

E9 `path_length` := "From t_1 to t_2: $s = \int_{t_1}^{t_2} \|\mathbf{v}(t)\|\, dt$."

E10 `point_particle` := "An idealized object whose size and internal structure are neglected; its motion in a reference_frame is fully represented by a trajectory $\mathbf{r}(t)$."

E11 `straight_line` := "A set $\mathbf{r}(\lambda) = \mathbf{r}_0 + \lambda\mathbf{u}$ with fixed $\mathbf{r}_0$, nonzero direction $\mathbf{u}$, and $\lambda \in \mathbb{R}$."

1.2 Uniform motion intermediates (non-law)

E12 `uniform_motion_on_interval(I)` := "A point_particle is in uniform_motion_on_interval(I) if its velocity is the same at every time in I; i.e. $\forall t_1, t_2 \in I,\ \mathbf{v}(t_1) = \mathbf{v}(t_2)$."

E13 `uniform_motion_at_time(t_0)` := "A point_particle is in uniform_motion_at_time(t_0) if $\exists \varepsilon > 0$ such that $\forall t \in (t_0 - \varepsilon, t_0 + \varepsilon)$, $\mathbf{v}(t) = \mathbf{v}(t_0)$."

1.3 Newton I (triplet: uniform motion as bridge; no acceleration)

E14 `uniform_motion_on_interval(I)` := "The state of motion exhibited by a free_particle when described in an inertial_frame over the interval I."

E15 `inertial_frame` := "A reference_frame in which every free_particle exhibits uniform_motion_at_time(t) (equivalently: uniform_motion_on_interval(I) on any interval of free motion)."

E16 `free_particle` := "A point_particle that, when described in an inertial_frame, exhibits uniform_motion_at_time(t) (equivalently: uniform_motion_on_interval(I)) whenever treated as free."

1.4 Newton II Law

E17 `acceleration` := "For a point_particle described in an inertial_frame, acceleration equals net_force divided by inertial_mass: $\mathbf{a} = \mathbf{F}_{\texttt{net}}/\texttt{inertial_mass}$ (equivalently: $\mathbf{F}_{\texttt{net}} = \texttt{inertial_mass} \cdot \mathbf{a}$)."

E18 `net_force` := "The vector quantity satisfying `net_force` = `inertial_mass` · `acceleration` for the point_particle (in an inertial_frame)."

E19 `inertial_mass` := "The positive scalar coefficient m such that `net_force` = $m \cdot$ `acceleration` for the point_particle (in an inertial_frame)."

1.5 Force sum Law

E20 `net_force(p,t)` := "For any particle p and time t, net_force(p, t) is the vector-sum of all acting_force elements in interacting_forces_set(p, t) (equivalently: $\texttt{net_force}(p,t) = \sum_{F \in \texttt{interacting_forces_set}(p,t)} F$)."

E21 `interacting_forces_set(p,t)` := "The set of acting_force vectors that the modeler judges to be acting on particle p at time t (i.e. the force-contributions currently included for p at t)."

E22 `acting_force` := "A vector force-contribution F that appears as an element of interacting_forces_set(p, t) for some particle p and time t."

1.6 Meta-typing (triplet: raw classes and recognition-based instantiation)

E23 `raw_class` := "An abstract object-type."

E24 `raw_class` := "An abstract object-type specified by class_specific_features(C); an object x is treated as instance_of(x, C) when it is recognized as possessing the class_specific_features(C)."

E25 `class_specific_features(C)` := "The set/list of features that specify raw_class C; these features are whatever a modeler must recognize x as having in order to treat x as instance_of(x, C)."

E26 `instance_of(x,C)` := "x is treated as an instance of raw_class C when x is recognized as possessing class_specific_features(C)."

1.7 Action-Reaction Law(1st part of Newton III)

E26 `Acting force.` := "*Acting force* is a force–contribution that is either a *canonical_force(acting_object)* or a *reaction_force(acting_object)*."

E28 `Canonical force.` := "A *canonical_force(acting_object)* is an acting force designated as canonical; the corresponding *reaction_force(acting_object)* is the acting force defined as the negative of it."

E29 `Reaction force.` := "A *reaction_force(acting_object)* is an acting force defined as the negative of *canonical_force(acting_object)*."

1.8 Acting object Law

E30 `acting_object` := An *acting_object* is an object equipped with both an *activation_condition(acting_object)* and a corresponding *canonical_force(acting_object)*. (This is a class)

E31 `Activation condition.` := If a particle fulfills the *activation_condition(acting_object)* the *canonical_force(acting_object)* will be an acting-force on the particle.

E32 `Canonical force (of an acting object).` := The *canonical_force(acting_object)* is the acting-force of a particle that fulfills the *activation_condition(acting_object)*.

1.9 Non-law

E33 `paired_particle(acting_object)` := Each *acting_object* carries an associated particle, called its *paired_particle*. The acting object is considered to be located at the location of its paired particle .The particle may act as the object's COM.

E34 `mechanical_system` := An arbitrary subset of particles $\{p_1, \ldots, p_n\}$ selected by the observer for analysis.

E35 `interaction_candidate` := For a particle p in a mechanical system S at time t, an *interaction candidate* is any acting force in $interacting_forces_set(p, t)$.

1.10 Mechanical Closure Law (2nd Part of Newton III)

E36 `interaction_candidate` := An *interaction candidate* is an elementary force-unit, every one of which must be able to be placed in an *interaction pair* for the mechanical system to be a *mechanically closed system*.

E37 `interaction_pair` := An *interaction pair* is the form into which *interaction candidates* are placed in a *mechanically closed system*.

E38 `mechanically_closed_system` := A *mechanically closed system* is a *mechanical system* in which every *interaction candidate* can be placed in an *interaction pair*.

1.11 Non-law

E39 `interaction_pair` := Two *interaction candidates* form an *interaction pair* when they correspond to the same *acting_object*, one being the *canonical_force* associated with that acting object and the other being the *reaction_force* associated with that acting object, acting on the object's *paired_particle*.

E40 `mechanically_closed_system` := A *mechanically closed system* is a *mechanical system* whose future behaviour is determined by the equations produced from the interactions internal to it.

2 Design 2

2.1 Kinematics (Non-law)

The entries in this scaffold express standard kinematic notions from introductory classical mechanics, cross-checked against sources such as Kleppner–Kolenkow and Taylor, but rewritten to satisfy VD entry-log dependency order. In particular, the early non-law entries are intended to capture familiar pre-law notions—time, reference frame, point particle, position, displacement, velocity, speed, acceleration, trajectory, straight line, path length, relative position, and relative velocity—in a form that uses only residue and earlier headwords. The conceptual content is source-backed; the phrasing and ordering are ours.

The Newton I and Newton II blocks that follow should be read somewhat differently. Their underlying physics is standard Newtonian mechanics, but their presentation as chimney, triplet, and wall entries belongs to the VD architecture rather than to any single textbook. Thus the scaffold entries are the ones most directly defended by the cited mechanics sources, while the later law-units are best understood as VD-native reorganisations of standard material.

E1 `time` := A real parameter $t \in \mathbb{R}$ used to order events; differences $\Delta t = t_2 - t_1$ are durations.

E2 `reference_frame` := A convention for assigning spatial coordinates and a time coordinate to events: an origin, a set of basis vectors, and a clock.

E3 `point_particle` := An idealised object whose spatial extent and internal structure are neglected.

E4 `position` := The location of a point particle at time t in a reference frame, represented by a vector $r(t) \in \mathbb{R}^3$.

E5 `displacement` := The change in position: $\Delta r = r(t_2) - r(t_1)$.

E6 `velocity` := $v(t) = \dfrac{dr}{dt}$.

E7 `speed` := The scalar magnitude of velocity: $\text{speed}(t) = \|v(t)\|$.

E8 `acceleration` := $a(t) = \dfrac{dv}{dt} = \dfrac{d^2r}{dt^2}$.

E9 `trajectory` := The map $t \mapsto r(t)$ giving the positions of a point particle over some interval $I \subseteq \mathbb{R}$.

E10 `straight_line` := A set of points $\{r_0 + \lambda u : \lambda \in \mathbb{R}\}$ for fixed $r_0 \in \mathbb{R}^3$ and nonzero direction $u \in \mathbb{R}^3$.

E11 `path_length` := For a differentiable trajectory from t_1 to t_2, the arc length is $s = \int_{t_1}^{t_2} \|v(t)\|\, dt$.

E12 `relative_position` := For two point particles in the same reference frame, the relative position is $r_{12}(t) = r_2(t) - r_1(t)$.

E13 `relative_velocity` := For two point particles in the same reference frame, the relative velocity is $v_{12}(t) = v_2(t) - v_1(t) = \dfrac{dr_{12}}{dt}$.

E14 `uniform_motion` := The state of being either stationary or moving along a straight line with constant speed. Equivalently, a point particle is in uniform motion on an interval I if its velocity is constant throughout I; i.e. $\forall t_1, t_2 \in I,\ v(t_1) = v(t_2)$. A point particle is in uniform motion at time t_0 if $\exists \varepsilon > 0$ such that it is in uniform motion on $(t_0 - \varepsilon,\ t_0 + \varepsilon)$.

2.2 Newton's 1st Law

E15 `uniform_motion` := The state of motion exhibited by a free particle when described in an inertial frame.

E16 `inertial_frame` := A reference frame in which every free particle exhibits uniform motion.

E17 `free_particle` := A point particle that, when described in an inertial frame, exhibits uniform motion.

2.3 Non-law

E18 `inertial_frame` := A reference frame standard for Newtonian analysis of motion; historically, a Galilean reference frame.

E19 `free_particle` := A point particle not subject to external influence relevant to its motion.

E20 `inertial_acceleration` := The acceleration of a point particle as measured in an inertial frame.

2.4 Newton's 2nd Law

E21 `inertial_acceleration` := For a point particle, inertial acceleration equals net force divided by inertial mass: $a = F_{\text{net}}/m$.

E22 `net_force` := The vector quantity satisfying $F_{\text{net}} = m \cdot a$ for a point particle, where a is inertial acceleration.

E23 `inertial_mass` := The positive scalar coefficient m such that $F_{\text{net}} = m \cdot a$ for a point particle, where a is inertial acceleration.

2.5 Force sum Law

E24 `net_force(p,t)` := "For any particle p and time t, net_force(p,t) is the vector-sum of all acting_force elements in interacting_forces_set(p,t) (equivalently: `net_force`$(p,t) = \sum_{F \in \texttt{interacting_forces_set}(p,t)} F$)."

E25 `interacting_forces_set(p,t)` := "The set of acting_force vectors that the modeler judges to be acting on particle p at time t (i.e. the force-contributions currently included for p at t)."

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E44 `mechanically_closed_system` := A *mechanically closed system* is a *mechanical system* whose future behaviour is determined by the equations produced from the interactions internal to it.

References

References

[1] Daniel Kleppner and Robert Kolenkow, *An Introduction to Mechanics*, 2nd ed., Cambridge University Press, 2014.

[2] John R. Taylor, *Classical Mechanics*, University Science Books, 2005.

[3] Isaac Newton, *Philosophiæ Naturalis Principia Mathematica*.